No Fault Found — Routine Preventive Check, Conveyor Motor B

Asset: AST_MTR_002 (Motor) | Bearing: BRG_003 (SKF6310)
Fault Mode: healthy | ISO Stage: No Stage (Healthy/Sensor)
Learned Case — Knowledge Agent Retrieval Document | DRO Baseline
CASE_005 | Outcome: PREVENTIVE | Opened: 2026-05-10 | Closed: 2026-05-10

1. Detection Summary

Detection Date	2026-05-10
Detected By	monitoring_agent (routine cycle — no anomaly raised)
Anomaly Score	0.21 (below alert threshold)
vib_rms_mm_s	1.9 mm/s (baseline mean: 1.8 mm/s — within ±1σ)
kurtosis	2.1 (well below fault threshold 5.0)
temp_c	52°C (baseline mean: 52°C — no deviation)
bpfo_energy	0.6× (well below stage_1 multiple 1.5×)
signal_quality_score	1.00 (perfect signal)
last_lubrication_date	2026-05-01 — 9 days before check, within interval

2. Agent Diagnosis

Primary Diagnosis	healthy — no fault
Confidence	0.92 (high confidence of no fault)
Reasoning	All signals within ±1σ of baseline. No threshold crossed. No trend indicating developing fault. Lubrication is current.
Differential	None required — healthy classification is unambiguous at this score.

3. Action Taken

Recommended Action	Routine preventive maintenance — grease top-up and inspection only
Window Selected	Next available planned stop on LINE_001
Parts Required	PART_004 (Synthetic Grease, small top-up quantity, BIN_L1)
Crew Assigned	K. Mensah (TECH_002)
LOTO Applied	LOTO_005 — MCC-13, Valve-5 (is_current: true)
Work Order	WO_005 — priority LOW

4. Findings at Inspection / Teardown

Visual inspection of motor exterior: clean, no leaks, fasteners tight. Grease nipple accessed for top-up. Existing grease at purge was light amber, clean, smooth texture — no contamination or discolouration. Added 35g PART_004. No bearing housing opened. No mechanical disassembly required. BRG_004 (NDE) was not accessed on this visit — visual condition of motor exterior gave no indication of NDE bearing issues.

5. Outcome

Post-Maintenance vib_rms	1.7 mm/s (QA pass: ≤ 2.5 mm/s) ✓
Post-Maintenance temp_c	51°C (QA pass: ≤ 60°C) ✓
Kurtosis at 30 min	2.1 (QA pass: ≤ 2.5) ✓
QA Result	PASS — healthy baseline confirmed
Actual Duration	2 hours
Actual Cost	INR 2,500
Recommendation Followed	Yes

6. Root Cause Confirmed

Root Cause	no_fault_found
Notes	Motor is in healthy operating condition. No action beyond lubrication required.

7. Lessons Learned

- AST_MTR_002 consistently shows stable signals: 1.8–2.0 mm/s vib and 51–54°C. Any reading above 2.5 mm/s or above 60°C warrants increased monitoring frequency.
- Healthy preventive checks provide the most reliable baseline data for the Monitoring Agent — stable readings from this check strengthened the baseline for BRG_003.
- Post-lubrication baseline: 1.7 mm/s vib, 51°C. Use as reference for this motor.
- Note: LOTO_005 (new permit for AST_MTR_002) is the correct permit for this asset — not LOTO_001.